# Assignment 1 (20)

Deadline: 29 October 2025 11:00 PM

**Submission link:** https://forms.gle/6NsNaBMpeYsDZqK38

[You need to solve 2 problems from Category A and 2 problem from Category B.]

**Rule:**

Last digit of Student ID = Even -----→ solve 1, 4, 5.a, 7

Last digit of Student ID = Odd -----→ solve 2, 3, 5.b, 6

## Category: A (Mark: 4 each)

1) Suppose a line segment starts at (120, 23) and ends at (423, 428). If we draw this problem using the DDA algorithm, determine the number of times x will be increased. How many pixels will be needed to draw this line?
2) Which problem persists in the DDA algorithm and how can it be solved?
3) Why is the value of y incremented/decremented by 1 instead of x, when the gradient is < -1 or > 1?
4) Do you think the Mid Point Line Drawing Algorithm is better than DDA? Provide reasoning.

## Category: B (Mark: 6 each)

5) Using the Digital Differential Analyzer (DDA) algorithm, find out the pixels between the lines whose endpoints are given:
   a. (5, 16) to (13, 10)
   b. (0, 0) to (4, 6)
6) There is a line with the equation, y = -5x - 20. The line segment starts from where it cuts the y-axis and ends at where it cuts the x-axis. Find the first 8 pixels using the Midpoint Line Drawing Algorithm.
7) Suppose a line segment has the equation $\frac{x}{7} - \frac{y}{12} = 5$, and the starting point is at y = 12 and the ending point is at y = 0. Find the first 6 pixels using the Midpoint Line Algorithm.